Lab Experiment 8: Change Passwords and Adjust Password Aging for Local User Accounts

Objective

To learn how to **change passwords** and **set password aging policies** for local user accounts in Linux using simple administrative commands.

Background Concept

In Linux, each user has a **password** stored in an encrypted form inside the `/etc/shadow` file. The **password aging feature** helps the administrator enforce rules like:

- How long a password remains valid
- When a user must change it
- Minimum days between password changes
- Warning before expiry

This ensures **security and discipline** in user account management.

Basic Terms

Term	Meaning
Password Aging	Controlling how long a password remains valid
/etc/shadow	File where encrypted passwords and password aging info are stored
passwd command	Used to change or set a user password
chage command	Used to view and modify password aging details

Step-by-Step Procedure

Step 1: Login as Root or Use sudo

To perform administrative actions, switch to root or use `sudo`:

```
sudo su
```

Step 2: Create a new User

Create a new user named student1::

```
sudo useradd -m student1
```

Set a password for the user:

```
sudo passwd student1
```

Step 3: View Password Aging Information

Use the `chage -l` command to display current aging settings for a user:

```
sudo chage -l student1
```

You'll see output like:

```
Last password change                                    : Oct 26,
    2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change           : 0
Maximum number of days between password change           : 99999
Number of days of warning before password expires        : 7
```

Step 4: Set Password Aging Rules

Task	Command	Example
Set password to expire after certain days	`chage -M <days> username`	`sudo chage -M 30 student1`
Set minimum days before user can change password again	`chage -m <days> username`	`sudo chage -m 2 student1`
Set warning days before password expires	`chage -W <days> username`	`sudo chage -W 5 student1`
Set account expiry date	`chage -E <YYYY-MM-DD> username`	`sudo chage -E 2025-12-31 student1`

Step 5: Verify the Updated Policy

After making changes:

```
sudo chage -l student1
```

Check whether the **Maximum**, **Minimum**, and **Warning** values are updated.

Common Administrative Tasks (for Practice)

Encourage students to try these tasks:

1. View password aging of your account (`chage -l <username>`)
2. Set your password to expire after 15 days.
3. Change another user's password.
4. Set warning 3 days before password expires.
5. Disable password expiration (set to "never").
6. Set account expiry date for testing purpose.
7. Attempt to log in after password expiry to observe system behavior.

Summary

- `passwd` → used to **set or change** passwords
- `chage` → used to **view or modify** password aging details
- `/etc/shadow` → stores all password and aging information securely

Learning Outcome

By the end of this experiment, students will be able to:

- Change user passwords safely.
- Understand password aging policies.
- Configure password expiry and warning systems.
- Maintain user account security through controlled password management.